35021C16	JAVA PROGRAMMING	Category	L	T	P	Credit
		CC	3	0	0	3

PREAMBLE

This course of study builds on the skills gained by students in Java Fundamentals and helps to advance Java programming skills. Students will design object-oriented applications with Java and will create Java programs using hands- on, engaging activities.

PREREQUISITE

NIL

COURSEOBJECTIVES

1	Understand fundamentals of programming such as variables,conditional and iterative execution,methods,etc.
2	Understand fundamentals of object-oriented programming in Java, including defining classes,invoking methods, Using class libraries,etc.
3	Be aware of the important topics and principles of software development.
4	Understand Event Handling and Swing Components.
5	Understand Generic Programming.

COURSEOUTCOMES

On successful completion of the course ,students will be able to

CO1.Knowledge of the structure and model of the Java programming language	Knowledge
CO2.Use the Java programming language for various programming technologies	Understand
CO3.Develop software in the Java programming language	Apply
CO4.Evaluate user requirements for software functionality required to decide whether the Java programming language can meet user requirements	Analyse
CO5.Choose an engineering approach to solving problems,Starting from the acquired knowledge of programming and knowledge of operating systems.	Evaluation

MAPPINGWITHPROGRAMMEOUTCOMESANDPROGRAMMESPECIFICOUTCOMES

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	S			S	S	M	S				S	S			
CO2	S		S		S			S	L	L		L			
CO3	S		M	L	S	M					L	L			
CO4	S		S	M	S		S				S	M			
CO5	S		S	M	S		M				S	M			

S-Strong;M-Medium;L-Low

SYLLABUS

BASICS OF JAVA

Object oriented programming concepts – objects – classes – methods and messages – abstraction and encapsulation –inheritance–abstract classes–polymorphism.-ObjectsandclassesinJava–defining classes–methods- access specifiers–static members –constructors–finalize method.

ARRAYS, OPERATORS, STRINGS &OBJECTS

Arrays - Operators: Arithmetic Operators, The Bit wise Operators, Relational Operators, Boolean Logical Operators, The Assignment Operator, The ? Operator, Operator Precedence, Using Parentheses, Control Statements: Java's Selection Statements, Iteration Statements, Jump Statements–Strings-Packages–Java-Doc comments–-Inheritance–class hierarchy–polymorphism–dynamic binding–final keyword–abstract classes- The Object class–Reflection–interfaces–object cloning–inner classes–proxies.

EVENTS & GRAPHICS PROGRAMMING

I/Streams- Filter and pipe streams– Byte Code interpretation- Basics of event handling – event handlers–adapter classes–actions–mouse events–AWTeventhierarchy–Graphics programming–Frame–Components–workingwith2Dshapes.

SWING & GENERIC PROGRAMMING, APPLETS

Introduction to Swing – Model-View-Controller design pattern – buttons – layout management – Swing Components –exception handling – exception hierarchy – throwing and catching exceptions - Motivation for generic programming –generic classes–generic methods–genericcodeandvirtualmachine–inheritanceandgenerics– reflectionandgenerics. Applets and HTML- Security Issues, Applets and Applications, passing parameters to applets. Creating a Swing Applet.

THREADS & SOCKET PROGRAMMING

Multi-threaded programming–interrupting threads–thread states–thread properties–thread synchronization– Executors–synchronizes–Socket Programming–UDP Datagram–IntroductiontoJavaBeans.

TEXTBOOKS:

1. CayS.HorstmannandGaryCornell,"CoreJava:VolumeI–Fundamentals",Ninth Edition, Sun MicrosystemsPress,2013.
2. ElliotteRustyHarold,"JavaNetworkProgramming",O"Reillypublishers,2000(UNITII).
3. EdRoman,"MasteringEnterpriseJavaBeans",JohnWiley&SonsInc.,1999(UNITIIIandUNITV).

REFERENCES:

1. K.ArnoldandJ.Gosling,"TheJAVAprogramminglanguage",Thirdedition,PearsonEducation,2008.
2. TimothyBudd,"Understanding Object-orientedprogrammingwithJava",UpdatedEdition,PearsonEducation,2000.
3. C.ThomasWu,"AnintroductiontoObject-orientedprogrammingwithJava",FourthEdition,TataMcGraw-HillPublishingcompanyLtd.,2006.

COURSEDESIGNERS

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